

Ridge Regression, Lasso Regression & ElasticNet Regression

Complete Notes with Detailed Step-by-Step Explanation

(Har chhoti se chhoti concept ko Hinglish mein, ekdum shuru se samjhaya gaya hai)

1. Introduction — Yeh Topic Kya Hai?

Linear Regression seekhne ke baad sabse pehla sawaal aata hai — jab model training data pe to accha chalta hai lekin naye (test) data pe accuracy gir jaati hai, tab kya karein? Isi problem ko Overfitting kehte hain, aur isko control karne ke liye teen important techniques use hoti hain — Ridge Regression, Lasso Regression, aur ElasticNet Regression. Teeno ko ek saath Regularization Techniques kaha jaata hai, aur ye Linear Regression ki hi hyperparameter tuning ka part hain.

2. Linear Regression — Quick Recap (Foundation)

Linear Regression mein hum X (independent feature) aur Y (dependent/output) ke beech ek best-fit straight line dhoondte hain:

$$h\theta(x) = \theta_0 + \theta_1 x$$

Yahan θ_0 intercept hai aur θ_1 slope. Line kitni accha fit kar rahi hai, yeh check karne ke liye Cost Function use hota hai — Mean Squared Error (MSE):

$$J(\theta) = (1/2m) \times \sum [h\theta(x^{(i)}) - y^{(i)}]^2$$

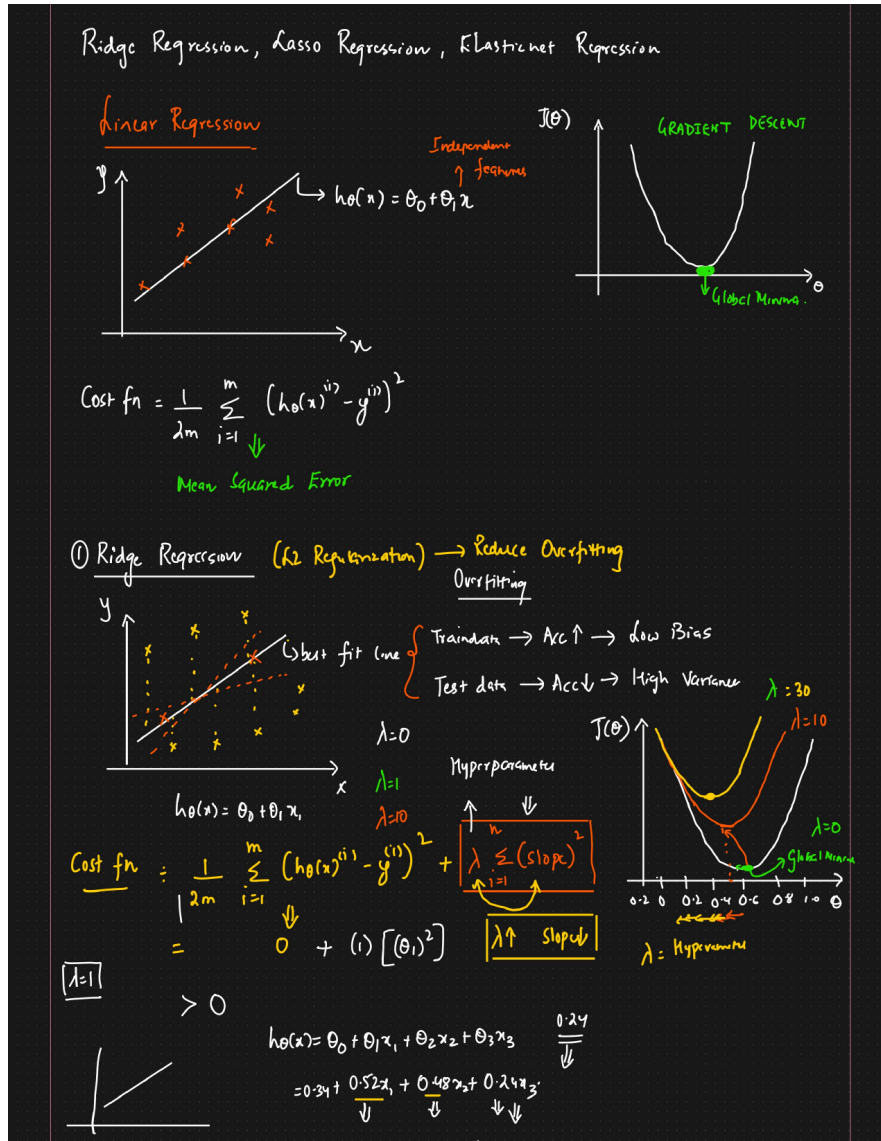
Is cost function ko minimum karne ke liye Gradient Descent algorithm chalta hai, jo dheere-dheere θ ki values update karke bowl-shaped $J(\theta)$ curve ke sabse neeche wale point — Global Minima — tak pahunchta hai.

3. Overfitting Ka Problem — Regularization Kyun Chahiye?

Jab model bahut complex ho jaata hai (bahut zyada features, ya bahut high-degree curve fit ho jaaye), to ek dikkat aati hai:

- Training data pe Accuracy $\uparrow$ (Low Bias)
- Test data pe Accuracy $\downarrow$ (High Variance)

Isi ko Overfitting kehte hain — model data ko "yaad" kar leta hai instead of asli pattern "seekhne" ke. Isko theek karne ke liye Ridge, Lasso aur ElasticNet — teeno regularization techniques use hoti hain, bas alag-alag tarike se.



Isme dikhaya gaya hai: Linear Regression recap (Hypothesis Function, Cost Function, Gradient Descent), Overfitting ka concept (Training vs Test accuracy), aur Ridge Regression ki shuruaat (Cost Function + L2 penalty term).

4. Ridge Regression (L2 Regularization)

Ridge, normal Cost Function mein ek extra penalty term jodta hai — jo coefficients (slopes) ke square ka sum hota hai:

$$J(\theta) = (1/2m) \times \sum [h\theta(x^{(i)}) - y^{(i)}]^2 + \lambda \times \sum (\text{slope})^2$$

Yahan λ (lambda) ek hyperparameter hai — matlab hum khud set karte hain, model khud nahi seekhta.

λ Ka Effect Samjho:

- $\lambda = 0 \rightarrow$ koi penalty nahi, ye simple Linear Regression jaisa hi behave karta hai
- λ badhta jaaye (1, 10, 30...) $\rightarrow$ penalty badhti jaati hai, cost curve "tang" hota jaata hai, aur coefficients (θ) chhote hote jaate hain
- Matlab: $\lambda \uparrow \Rightarrow$ Slope $\downarrow$ — jitna zyada λ , model utna zyada "simple" banega, taaki overfitting kam ho

Example (Notes se):

Before Ridge: $h\theta(x) = 0.34 + 0.52x_1 + 0.48x_2 + 0.24x_3$

After Ridge: $h\theta(x) = 0.34 + 0.40x_1 + 0.38x_2 + 0.14x_3$

Important: Ridge coefficients ko shrink karta hai, lekin kabhi bhi bilkul zero nahi karta. Isliye Ridge saare features ko model mein rakhta hai, bas unka weight/importance kam kar deta hai. Isi wajah se Ridge ka main kaam hai — Reduce Overfitting.

5. Lasso Regression (L1 Regularization)

Lasso bhi bilkul Ridge jaisa hi kaam karta hai, bas penalty term mein square ki jagah coefficients ka absolute value (mod) use hota hai:

$$J(\theta) = (1/2m) \times \sum [h\theta(x^{(i)}) - y^{(i)}]^2 + \lambda \times \sum |\text{slope}|$$

Ridge aur Lasso Ka Sabse Bada Difference:

- Ridge $\rightarrow$ slope² (square) $\rightarrow$ coefficients shrink hote hain, but zero nahi hote
- Lasso $\rightarrow$ |slope| (mod/absolute) $\rightarrow$ coefficients zero tak bhi ja sakte hain

Example (Notes se):

Before Lasso: $h\theta(x) = 0.52 + 0.65x_1 + 0.72x_2 + 0.34x_3 + 0.12x_4$

After Lasso: $h\theta(x) = 0.52 + 0.81x_1 + 0.60x_2 + 0.14x_3 + 0 \times x_4$

Dekho — x_4 ka coefficient bilkul 0 ho gaya, matlab Lasso ne us feature ko poori tarah "drop" kar diya, jaise wo model mein hai hi nahi. Isi wajah se Lasso ka main use hota hai Feature Selection — jo features model ki prediction mein zyada contribute nahi karte, unhe Lasso automatically zero karke hata deta hai.

6. ElasticNet Regression

ElasticNet simple si baat hai — Ridge + Lasso dono ka combination. Isme dono penalty terms ek saath cost function mein add ho jaate hain:

$$J(\theta) = (1/2m) \times \sum [h\theta(x^{(i)}) - y^{(i)}]^2 + \lambda_1 \times \sum (\text{slope})^2 + \lambda_2 \times \sum |\text{slope}|$$

- Pehla term ($\lambda_1 \sum \text{slope}^2$) $\rightarrow$ Ridge wala part $\rightarrow$ Overfitting reduce karta hai
- Dusra term ($\lambda_2 \sum |\text{slope}|$) $\rightarrow$ Lasso wala part $\rightarrow$ Feature Selection karta hai

Matlab ElasticNet dono ke fayde ek saath deta hai — overfitting bhi kam hoti hai aur useless features bhi automatically zero ho jaate hain. Isme do hyperparameters (λ_1 aur λ_2) hote hain, jinhe tune karna padta hai.

$$= 0.34 + 0.40x_1 + 0.38x_2 + \boxed{0.14x_3}$$

② Lasso Regression (L_1 Regularization) $\rightarrow$ Feature Selection

Cost fn $= \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2 + \lambda \sum_{i=1}^n |\text{slope}|$

$h_{\theta}(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 + \theta_4 x_4$
 $h_{\theta}(x) = 0.52 + 0.65x_1 + 0.72x_2 + 0.34x_3 + \boxed{0.12x_4}$

$\downarrow$ Lasso Regression $\downarrow$ Feature Selection
 $= 0.52 + 0.51x_1 + 0.60x_2 + 0.14x_3 + \boxed{0x_4}$

③ ElasticNet Regression

$\rightarrow$ ① Reduce Overfitting
 $\rightarrow$ ② Feature Selection

Cost fn $= \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2 + \lambda_1 \sum_{i=1}^n |\text{slope}|^2 + \lambda_2 \sum_{i=1}^n |\text{slope}|$

$\downarrow$ Reduce Overfitting $\downarrow$ Feature Selection

$\left\{ \begin{array}{l} \text{Hyperparameter Tuning the} \\ \text{Linear Regression} \end{array} \right\}$

Isme dikhaya gaya hai: Lasso Regression ka Cost Function (L_1 penalty) with worked example (coefficient zero ho jaana), ElasticNet Regression ka Cost Function (Ridge + Lasso combined), aur note ki ye poora topic Linear Regression ki Hyperparameter Tuning ka hissa hai.

7. Quick Summary Table (Revision Ke Liye)

	Ridge (L2)	Lasso (L1)	ElasticNet
Penalty Term	$\lambda \times \sum \theta^2$	$\lambda \times \sum \theta $	Dono (λ_1, λ_2)
Coefficient zero hota hai?	Nahi, sirf shrink	Haan, ho sakta hai	Haan
Main Kaam	Overfitting kam karna	Feature Selection	Overfitting + Feature Selection

Hyperparameter	λ	λ	λ_1, λ_2
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Ek line mein yaad rakhna: λ ek hyperparameter hai jo control karta hai ki penalty kitni strong ho — agar $\lambda = 0$ ho, sab normal Linear Regression ban jaata hai; jitna λ badhega, model utna hi "simple" hota jayega.